

Beijing Summer School
Appendix 1: Stochastic Calculus

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“Life is good for only two things, discovering mathematics and teaching mathematics.”

–Siméon Denis Poisson (1781-1840).

Lecture Goals

- Introduce Brownian motion and Poisson processes
- Introduce the stochastic integral
- Introduce Itô's lemma
- Introduce stochastic “differential” algebraic rules

References on Stochastic Calculus

- Introductory Texts:
[Øksendal, 2003], [Klebaner, 2012], and [Hanson, 2007]
- More advanced texts on Brownian motion:
[Karatzas and Shreve, 1998], [Schilling and Partzsch, 2014],
[Liptser and Shiryaev, 1986], [Shiryaev, 1999] or [Protter, 1992]
- More advanced texts on Lévy processes (jump processes):
[Applebaum, 2009]

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Definitions and Notation About Stochastic Processes

- The triple $(\Omega, \mathcal{F}, P)$ is a **probability space**. Elements of Ω are called **outcomes**, elements of $\mathcal{F}$ are called **events**, $P(A)$ is **probability** of an event $A \in \mathcal{F}$ occurring.
- A stochastic process $\{X_t : t \in \mathcal{T}\}$ is family of random variables with values at each time t .
- A **Filtration** $\{F_t : t \in \mathcal{T}\}$ (generated by X) is the “information” obtained by observing the history of a stochastic process $\{X_t : 0 \leq s \leq t\}$.
- More formal introduction to stochastic processes in my notes.

Markov property

Definition

A stochastic process has the **Markov property** if, for all $s, t \geq 0$ and for every Borel set $H \in \mathcal{B}$ on $\mathcal{B}^k$, the following holds:

$$P(X_{t+s} \in H | \mathcal{F}_t) = P(X_{t+s} \in H | X_t)$$

- Conceptually, this says that a Markov process is “memoryless”.

p-Variation

Definition

The set of points $\mathcal{P} = \{t_0, \dots, t_n\}$ with $0 = t_0 < t_1 < \dots < t_n = t$ is a **partition of the interval** $[0, t]$. The **norm of the partition** is $l(\mathcal{P}) = \max |t_j - t_{j-1}|$

Definition

Let $X : \Omega \times \mathcal{T} \rightarrow \mathbb{R}$ be a stochastic process.

- For $p > 0$, define the **p -th variation process** of X_t as the following (where the limit is in probability):

$$\langle X, X \rangle_t^p(\omega) = \lim_{l(\mathcal{P}) \rightarrow 0} \sum_{j=0}^{n-1} \left| X_{t_{j+1}}(\omega) - X_{t_j}(\omega) \right|^p$$

- For $p = 1$, this is the **(total) variation process**
- For $p = 2$, this is the **quadratic variation process**.

Interpretation of Variation

- $\langle X, X \rangle_t^p(\omega)$ is a measure of how much a stochastic process “moves around”
- $\langle X, X \rangle_t^p(\omega)$ is weakly decreasing in p
(i.e. Easier to get $\langle X, X \rangle_t^p(\omega) < \infty$ as $\uparrow p$)
- Why? $\uparrow p$ “dampens” $\left| X_{t_{j+1}}(\omega) - X_{t_j}(\omega) \right|^p$ as $\left| X_{t_{j+1}}(\omega) - X_{t_j}(\omega) \right| \rightarrow 0$

Quadratic Variation and Covariation

- Quadratic variation is particularly important in stochastic calculus
- Special notation for quadratic variation: $[X]_t$ or $[X, X]_t$
- The **quadratic covariation between two stochastic processes** X, Y is:

$$[X, Y]_t(\omega) := \lim_{l(\mathcal{P}) \rightarrow 0} \sum_{j=0}^{n-1} (X_{t_{j+1}}(\omega) - X_{t_j}(\omega))(Y_{t_{j+1}}(\omega) - Y_{t_j}(\omega))$$

- If X is a continuous stochastic process and Y is a stochastic process with finite variation, then:

$$[X, Y]_t(\omega) = 0$$

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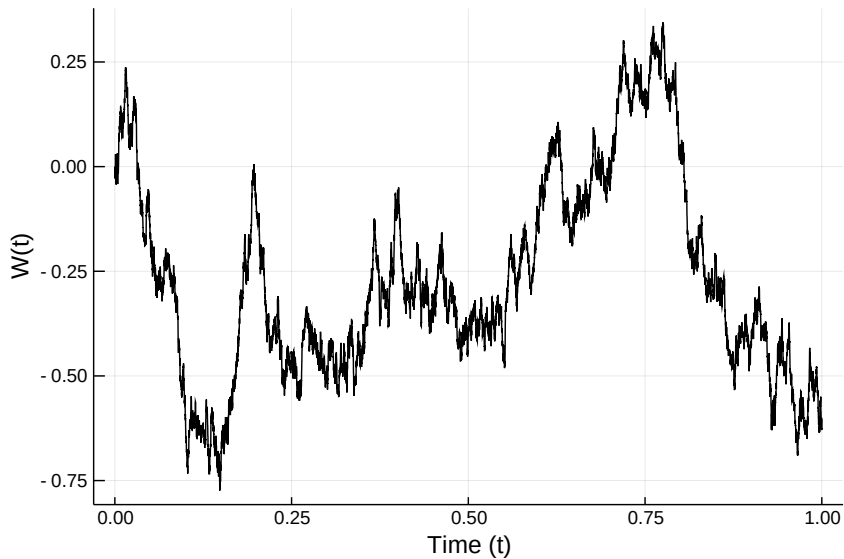
Brownian motion

Definition:

A **k -dimensional Brownian motion** is a stochastic process W on $\mathbb{R}^k$ s.t.

1. $W_0(\omega) = 0$, a.s.
2. (*Normal increments*): $\forall s, t \in \mathcal{T}$ for which $s \leq t$, the difference $W_t - W_s \sim N(0, (t - s)I_k)$
3. (*Independent increments*): For all $t_0 < t_1 < t_2 < \dots t_n \in \mathcal{T}$, the random variables $W_{t_j} - W_{t_{j-1}}$, $j \in \{1, \dots, n\}$ are independent.
4. (*Continuity of sample paths*): $t \mapsto W_t(\omega)$ is continuous for all ω .

Sample Paths



Properties of Brownian Motions

- Brownian motion has Markov property: $\mathbb{P}(X_{t+s} \leq x | \mathcal{F}_t) = \mathbb{P}(X_{t+s} \leq x | X_t)$
- Brownian motion is a martingale: $\mathbb{E}(X_{t+s} | \mathcal{F}_t) = X_t$
- A Brownian motion path, $\{W_t : 0 \leq t \leq T\}$, has the following properties:
 1. is a continuous function of t (by the definition);
 2. is not monotone in any interval, no matter how small the interval is;
 3. is of infinite variation (over any arbitrary small interval) and finite quadratic variation:

$$\langle W, W \rangle_t^1(\omega) = \infty, \quad \forall t > 0$$

$$[W, W]_t(\omega) = \langle W, W \rangle_t^2(\omega) = t, \quad \forall t > 0, \quad a.s.$$

4. is not differentiable at any point.
- Breiman/Harrison: if X has continuous sample paths and X has i.i.d. increments, then X is a Brownian motion.

Q. What does infinite variation mean?

- A. Suppose you try to draw sample path of a process X over time interval $t \in [0, \Delta]$
- X has finite variation: for sufficiently small Δ , you can draw the line with finite ink
- X has infinite variation: no matter how small Δ gets, you will always need infinite ink to draw the line

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Poisson Process

Definition:

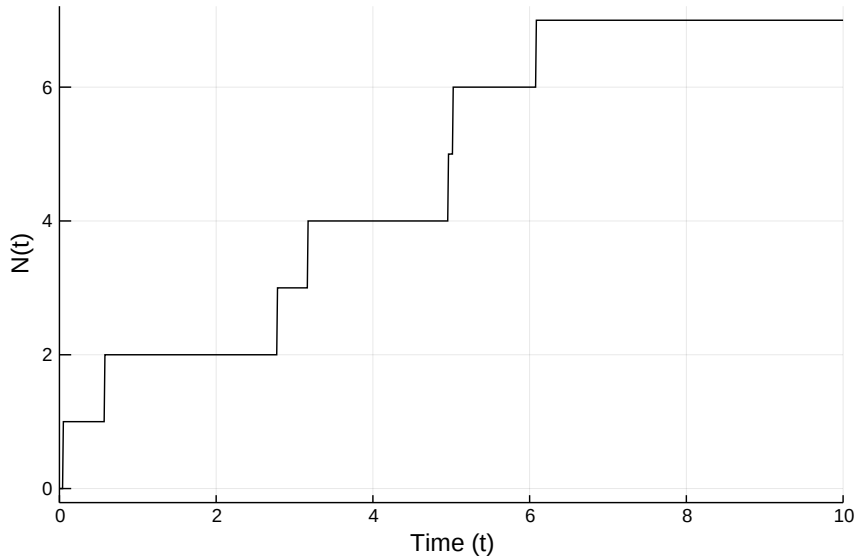
Let $\{\mathcal{F}_t\}$ be the history of realizations (called a filtration). A **(simple) Poisson process** with parameter $\lambda > 0$ is a stochastic process N satisfying the following properties:

- $N_0 = 0$, a.s.
- (*Stationary Poisson increments*): If $s < t$, then $N_t - N_s \sim \text{Pn}(\lambda(t - s))$. That is:

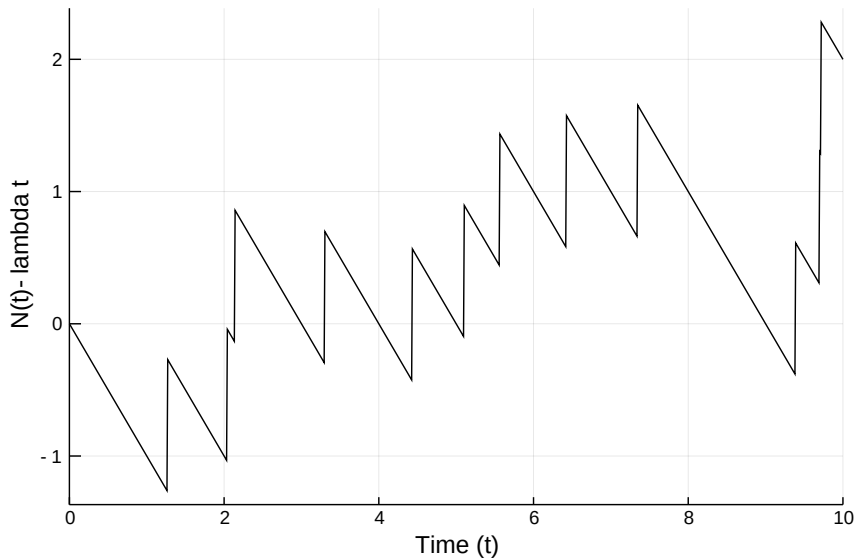
$$\mathbb{P}(N_t - N_s = k) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!} \quad (1)$$

- (*Independent Increments*): $P_t - P_s$ is independent of $\mathcal{F}_s$ for any $0 \leq s < t \leq T$.
- (*Right continuity*): The paths of P_t are right continuous with left limits.

Sample Path: N_t



Sample Path: $\bar{N}_t = N_t - \lambda t$



Interpretation of Poisson Processes

- N_t counts the number of unit sized jumps:

$$N_t = \sum_{k \geq 1} \mathbb{1}_{[T_k, \infty)}(t), \quad t \in \mathbb{R}_+$$

where $k \geq 1$, $\{T_k\}_{k \geq 1}$ is the increasing family of jump times of $\{N_t\}_{t \in \mathbb{R}_+}$ such that $\lim_{k \rightarrow \infty} T_k = +\infty$, and the indicator function records whether a jump happened at t

$$\mathbb{1}_{[T_k, \infty)}(t) = \begin{cases} 1, & \text{if } t \geq T_k, \\ 0, & \text{if } 0 \leq t < T_k, \end{cases}$$

- Once we “compensate” by subtracting λt from the Poisson process we get a stationary process. We call $N_t - \lambda t$ the **compensated Poisson process**.

Properties of Poisson Processes

1. For any k , the random inter-jump time $\tau_k := T_{k+1} - T_k$ (i.e. the waiting time between the k and $k + 1$ jumps) is exponentially distributed with parameter λ ;
2. The Poisson process has the markov property: $\mathbb{P}(N_{t+s} \leq x | \mathcal{F}_t) = \mathbb{P}(N_{t+s} \leq x | N_t)$
3. The Poisson process is not a martingale. However, $N_t - \lambda t$ is a martingale;
4. The Poisson process has finite variation and finite quadratic variation:

$$\begin{aligned}\langle N, N \rangle_t^1(\omega) &= N_t, \quad \forall t > 0 \\ [N, N]_t(\omega) &= \langle N, N \rangle_t^2(\omega) = N_t, \quad \forall t > 0, \quad a.s.\end{aligned}$$

and so:

$$[W, N]_t(\omega) = \langle W, N \rangle_t^2(\omega) = 0$$

5. For h small, $N_{t+h} - N_t$ behaves like a Bernoulli random variable with parameter λh . That is, a jump happens w.p. $h\lambda$ and does not happen w.p. $1 - h\lambda$.

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Goal

- We want to be able to define integrals of the form

$$I_t(X)(\omega) = \int_0^t X_s(\omega) dW_s(\omega), \quad W \text{ is Brownian motion}$$

and

$$I_t(X)(\omega) = \int_0^t X_s(\omega) dN_s(\omega), \quad N \text{ is Poisson process}$$

- What are the tools you already know?

Recall: Reimann Integration

- The **Reimann Integral** of function f over an interval $[a, b]$ is defined as:

$$\int_a^b f := \int_a^b f(t)dt := \lim_{\delta \rightarrow 0} \sum_i f(\xi_i^n) (t_i^n - t_{i-1}^n)$$

where the $\{t_i^n\}$ represent partitions of the interval and:

$$a = t_0^n < t_1^n < \dots < t_n^n = b,$$

$$\delta = \max_{1 \leq i \leq n} (t_i^n - t_{i-1}^n), \text{ and}$$

$$t_{i-1}^n \leq \xi_i^n \leq t_i^n$$

Recall: Stieltjes and Lebesgue Integration

- Function notation:
 - Let $G : \mathcal{T} \rightarrow \mathbb{R}$ be a function of finite variation.
 - Let $f : \mathcal{T} \rightarrow \mathbb{R}$ be a function
- The **Stieltjes Integral** of function f w.r.t. G over an interval $[a, b]$ is defined as:

$$\int_a^b f dG := \int_a^b f(t) dG(t) := \lim_{\delta \rightarrow 0} \sum_i f(\xi_i^n) (G(t_i^n) - G(t_{i-1}^n))$$

where the $\{t_i^n\}$ represent partitions of the interval and:

$$a = t_0^n < t_1^n < \dots < t_n^n = b,$$

$$\delta = \max_{1 \leq i \leq n} (t_i^n - t_{i-1}^n), \text{ and}$$

$$t_{i-1}^n \leq \xi_i^n \leq t_i^n$$

- The **Lebesgue Integral** “inverts” the partition to allow for complex G functions.

Stochastic Integration

- Our goal: define integrals of the form:

$$I_t(X)(\omega) = \int_0^t X_s(\omega) dW_s(\omega)$$

$$I_t(X)(\omega) = \int_0^t X_s(\omega) dN_s(\omega)$$

- Difficulty: the integral $I_t(X)(\omega)$ is now a random variable itself
- Approach 1: Define the integral pathwise (“a.s. convergence”)
 - For each ω , take the function $G(t)(\omega) := W_t(\omega)$ or $G(t)(\omega) := N_t(\omega)$
 - Then integrate with respect to the measure induced by G
 - This works for N because Poisson processes have finite variation
 - However, doesn’t work W because BM has infinite variation
- Approach 2: “new” integral! (“convergence in expectation”)
 - Takes the limit in expectation (rather than along each path)
 - Derivation details are beyond the scope of this course

Intuition*: Stochastic Integral of X_t w.r.t. W_t (i.e. $\int_0^t X_s(\omega)dW_s(\omega)$)

1. Consider **simple processes** that take on a discrete set of j different values:

$$\phi_t(\omega) = \phi_0(\omega)I_{\{t=0\}}(t) + \sum_{i=0}^{j-1} \phi_i(\omega)I_{(t_i, t_{i+1}]}(t), \quad t \in [0, T]$$

Define the integral for a simple process as:

$$\widehat{I}_t(\phi)(\omega) = \int_0^t \phi_s(\omega)dW_s(\omega) := \sum_{i=0}^{j-1} \phi_i(\omega)[W_{t_{i+1}}(\omega) - W_{t_i}(\omega)]$$

2. For a given stochastic process X , find an approximating sequence of simple processes $\{\phi^n\}$ such that:

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^t (\phi_s^n - X_s)^2 ds \right] \rightarrow 0$$

3. Define the stochastic integral as the limit of the simple integrals:

$$I_t(X)(\omega) = \int_0^t X_s(\omega)dW_s(\omega) = \lim_{n \rightarrow \infty} \widehat{I}_t(\phi^n)(\omega)$$

Key Properties of Stochastic Integrals

1. For Brownian motions (if X_t is square integrable, $\mathbb{E} \left[\int_0^T X_t^2 dt \right] < \infty$):

$$\mathbb{E} \left[\int_{\tau_0}^{\tau_1} X_t dW_t \mid \mathcal{F}_{\tau_0} \right] = 0, \quad \text{and}$$
$$\mathbb{E} \left[\int_0^{\tau_1} X_t dW_t \mid \mathcal{F}_{\tau_0} \right] = \int_0^{\tau_0} X_t dW_t, \quad 0 < \tau_0 < \tau_1$$

2. For Poisson processes (if X_t is square integrable, $\mathbb{E} \left[\int_0^T X_t^2 dt \right] < \infty$):

$$\mathbb{E} \left[\int_{\tau_0}^{\tau_1} X_{s-} dN_s \mid \mathcal{F}_{\tau_0} \right] = \int_{\tau_0}^{\tau_1} \mathbb{E} [X_s \mid \mathcal{F}_{\tau_0}] \lambda ds$$

Covariation of Stochastic Integrals

- Let X and Y be stochastic processes. Let I^X and I^Y denote the stochastic processes defined by the integrals:

$$I_t^X := \int_0^t H_s dX_s, \quad I_t^Y := \int_0^t K_s dY_s$$

- Then:

- The quadratic covariation:

$$[I^X, I^Y] = \left[\int_0^\cdot H_s dX_s, \int_0^\cdot K_s dY_s \right]_t = \int_0^t H_s K_s d[X, Y]_s$$

- If the integrals are both w.r.t. the stochastic process X :

$$[I^X, I^Y] = \left[\int_0^\cdot H_s dX_s, \int_0^\cdot K_s dX_s \right]_t = \int_0^t H_s K_s d[X, X]_s$$

- If only one process is an integral:

$$\left[\int_0^\cdot H_s dX_s, Y \right]_t = \left[\int_0^\cdot H_s dX_s, \int_0^\cdot 1 dY_s \right]_t = \int_0^t H_s d[X, Y]_s$$

Examples

- $\left[\int_0^t \sigma_s^X dW_s, \int_0^t \sigma_s^Y dW_s \right]_t = \int_0^t \sigma_s^X \sigma_s^Y d[W, W]_s = \int_0^t \sigma_s^X \sigma_s^Y ds,$
(where I have used that $[W, W]_t = t$),
- $\left[\int_0^t \sigma_s dW_s, \int_0^t \sigma_s dW_s \right]_t = \int_0^t \sigma_s^2 d[W, W]_s = \int_0^t \sigma_s^2 ds,$
- $\left[\int_0^t \sigma_s dW_s, \int_0^t \mu_s ds \right]_t = 0$, since the integral of a Brownian motion process is continuous and the second integral is not stochastic and so has finite variation.
- $\left[\int_0^t \gamma_s^X dN_s, \int_0^t \gamma_s^Y dN_s \right]_t = \int_0^t \gamma_s^X \gamma_s^Y d[N, N]_s = \int_0^t \gamma_s^X \gamma_s^Y dN_s,$
(where I have used that $[N, N]_t = N_t$).

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Itô Process

- An **Itô process** takes the form:

$$X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s + \int_0^t \gamma_s dN_s, \quad 0 \leq t \leq T$$

where:

- Y_0 is $\mathcal{F}_0$ -measurable,
 - The processes μ_t , σ_t and γ_t are $\mathcal{F}_t$ -adapted and satisfy $\mathbb{E}[\int_0^T |\mu_s| ds] < \infty$, $\mathbb{E}[\int_0^T \sigma_s^2 ds] < \infty$ and $\mathbb{E}[\int_0^T |\gamma_s| ds] < \infty$
 - μ_s , σ_s and γ_s may depend upon X_s .
- Say: Itô process X has the **stochastic differential** on $[0, T]$:

$$dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t, \quad 0 \leq t \leq T$$

- Technical aside: Differentials are defined by the integral equation.

Quadratic Covariation of Itô Processes

- If $dX_t = \mu_t dt + \sigma_t dW_t$, then:

$$\begin{aligned}[X, X]_t &= \int_0^t \sigma_s^2 ds \\ \Rightarrow d[X, X]_t &= \sigma_t^2 dt\end{aligned}$$

- If $dX_t = \mu_t dt + \sigma_t dW_t + \gamma_t dN_t$, then:

$$\begin{aligned}[X, X]_t &= \int_0^t \sigma_s^2 ds + \int_0^t \gamma_s^2 dN_s \\ \Rightarrow d[X, X]_t &= \sigma_t^2 dt + \gamma_t^2 dN_t\end{aligned}$$

- Proof: use earlier results and linearity of quadratic covariation

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Itô's lemma (Univariate)

Lemma (Univariate Case, No Jumps, Integral Form)

Suppose that $f(x)$ is twice differentiable (i.e. a C^2 function) and X_t satisfies the following stochastic differential for $0 \leq t \leq T$:

$$dX_t = \mu_t dt + \sigma_t dW_t$$

Then,

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d[X, X]_s \\ &= f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) \sigma_s^2 ds \\ &= f(X_0) + \int_0^t (f'(X_s) \mu_s + \frac{1}{2} f''(X_s) \sigma_s^2) ds + \int_0^t f'(X_s) \sigma_s dW_s \end{aligned}$$

Itô's lemma (Univariate)

Lemma (Univariate Case, No Jumps. Differential Form)

Suppose that $f(x)$ is twice differentiable (i.e. a C^2 function) and X_t satisfies the following stochastic differential for $0 \leq t \leq T$:

$$dX_t = \mu_t dt + \sigma_t dW_t$$

Then,

$$\begin{aligned} df(X_t) &= f'(X_t)dX_t + \frac{1}{2}f''(X_t)d[X, X]_t \\ &= f'(X_t)(\mu_t dt + \sigma_t dW_t) + \frac{1}{2}f''(X_t)\sigma_t^2 dt \\ &= (f'(X_t)\mu_t + \frac{1}{2}f''(X_t)\sigma_t^2)dt + f'(X_t)\sigma_t dW_t \end{aligned}$$

Itô's lemma (Univariate)

Lemma (Two Variable Case, No Jumps, Differential Form)

Suppose that $f(x, y) \in C^{2,2}$ and X_t and Y_t are Itô processes satisfying the following stochastic differential equations for $0 \leq t \leq T$:

$$dX_t = \mu_{X,t}dt + \sigma_{X,t}dW_t$$

$$dY_t = \mu_{Y,t}dt + \sigma_{Y,t}dW_t$$

Then,

$$\begin{aligned}df(X_t, Y_t) &= \partial_x f(X_t, Y_t)dX_t + \partial_y f(X_t, Y_t)dY_t + \partial_{xy}f(X_t, Y_t)d[X, Y]_t \\&\quad + \frac{1}{2}\partial_{xx}f(X_t, Y_t)d[X, X]_t + \frac{1}{2}\partial_{yy}f(X_t, Y_t)d[Y, Y]_t \\&= \partial_x f(X_t, Y_t)dX_t + \partial_y f(X_t, Y_t)dY_t + \partial_{xy}f(X_t, Y_t)\sigma_{X,t}\sigma_{Y,t}dt \\&\quad + \frac{1}{2}\partial_{xx}f(X_t, Y_t)\sigma_{X,t}^2dt + \frac{1}{2}\partial_{yy}f(X_t, Y_t)\sigma_{Y,t}^2dt\end{aligned}$$

Itô's lemma (Two Variables)

Lemma (Time Varying Univariate Case, No Jumps, Diff. Form)

Suppose that $f(x, t) \in C^{2,1}$ and X_t satisfies the stochastic differential for $0 \leq t \leq T$:

$$dX_t = \mu_t dt + \sigma_t dW_t$$

Then,

$$\begin{aligned} df(X_t) &= \partial_t f(X_t, t) dt + \partial_x f(X_t, t) dX_t + \frac{1}{2} \partial_{xx} f(X_t, t) d[X, X]_t \\ &= \partial_t f(X_t, t) dt + \partial_x f(X_t, t) (\mu_t dt + \sigma_t dW_t) + \frac{1}{2} \partial_{xx} f(X_t, t) \sigma_t^2 dt \\ &= \left(\partial_t f(X_t, t) + \partial_x f(X_t, t) \mu_t + \frac{1}{2} \partial_{xx} f(X_t, t) \sigma_t^2 \right) dt \\ &\quad + \partial_x f(X_t, t) \sigma_t dW_t \end{aligned}$$

Itô's lemma (Multivariate)

Lemma (Multivariate Case, No Jumps, Differential Form)

Suppose that $f(x_t^1, \dots, x_t^n) \in C^2$ for n variables and $\mathbf{X}_t$ is an n -dim. process:

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \sigma_t d\mathbf{B}_t, \quad \mathbf{X}_t \in \mathbb{R}^n, \boldsymbol{\mu}_t \in \mathbb{R}^n, \mathbf{B}_t \in \mathbb{R}^m, \sigma_t \in \mathbb{R}^{n,m}$$

Then:

$$\begin{aligned} df(X_t^1, \dots, X_t^n) &= \sum_{i=1}^n \frac{\partial}{\partial x_i} f(X_t^1, \dots, X_t^n) dX_t^i + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2}{\partial x_i \partial x_j} f(X_t^1, \dots, X_t^n) d[X^i, X^j]_t \\ &= D_x f(\mathbf{X}_t) \cdot d\mathbf{X}_t + \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n [D_x^2 f(\mathbf{X}_t)]_{i,j} [\sigma_t \sigma_t^T]_{i,j} dt \\ &= D_x f(\mathbf{X}_t) \cdot d\mathbf{X}_t + \frac{1}{2} \text{tr} \{ \sigma_t \sigma_t^T D_x^2 f(\mathbf{X}_t) \} dt \\ &= \left(D_x f(\mathbf{X}_t) \cdot \boldsymbol{\mu}_t + \frac{1}{2} \text{tr} \{ \sigma_t \sigma_t^T D_x^2 f(\mathbf{X}_t) \} \right) dt + D_x f(\mathbf{X}_t) \sigma_t dW_t \end{aligned}$$

Itô's lemma (with Jumps)

Lemma (Univariate Case, Jumps, Differential Form)

Suppose that $f(x)$ is a non-stochastic, C^1 function. Suppose that $X_t \in \mathbb{R}$ follows:

$$dX_t = \mu_t dt + \gamma_t dN_t$$

Then;

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t \partial_x f(X_s) \mu_s ds + \int_0^t (f(X_s) - f(X_{s-})) dN_s \\ &= f(X_0) + \int_0^t \partial_x f(X_s) \mu_s ds + \int_0^t (f(X_{s-} + \gamma_s) - f(X_{s-})) dN_s \end{aligned}$$

which in differential notation is:

$$\begin{aligned} df(X_t) &= \partial_x f(X_t) \mu_t dt + (f(X_t) - f(X_{t-})) dN_t \\ &= \partial_x f(X_t) \mu_t dt + (f(X_{t-} + \gamma_t) - f(X_{t-})) dN_t \end{aligned}$$

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Notational Convention

- The bivariate Itô's lemma implies the **stochastic integration by parts** formula:

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + d[X, Y]_t$$

- This motivates the common mathematical convention:

$$dX_t dY_t := d[X, Y]_t$$

so that the expression for $d(X_t Y_t)$ looks superficially like the “Taylor expansion”:

$$d(X_t Y_t) = X_t dY_t + Y_t dX_t + dX_t dY_t$$

- We can use our results on quadratic covariation to write down a set of algebraic rules for multiplying stochastic differentials.

Algebra Rules (in cross table)

	dt	$dW(t)$	$dN(t)$
dt	0	0	0
$dW(t)$	0	dt	0
$dN(t)$	0	0	dN_t

Example

- Suppose that:

$$dX_t = \mu_{X,t}dt + \sigma_{X,t}dW_t + \gamma_{X,t}dN_t$$

$$dY_t = \mu_{Y,t}dt + \sigma_{Y,t}dW_t + \gamma_{Y,t}dN_t$$

- Then:

$$\begin{aligned}dX_t dY_t &= (\mu_{X,t}dt + \sigma_{X,t}dW_t + \gamma_{X,t}dN_t)(\mu_{Y,t}dt + \sigma_{Y,t}dW_t + \gamma_{Y,t}dN_t) \\&= + \mu_{X,t}\mu_{Y,t}(dt)^2 + \mu_{X,t}\sigma_{Y,t}dtdW_t + \mu_{X,t}\gamma_{Y,t}dtdN_t \\&\quad + \sigma_{X,t}\mu_{Y,t}dW_tdt + \sigma_{X,t}\sigma_{Y,t}(dW_t)^2 + \sigma_{X,t}\gamma_{Y,t}dW_tdN_t \\&\quad + \gamma_{X,t}\mu_{Y,t}dN_tdt + \gamma_{X,t}\sigma_{Y,t}dN_tdW_t + \gamma_{X,t}\gamma_{Y,t}(dN_t)^2 \\&= \sigma_{X,t}\sigma_{Y,t}dt + \gamma_{X,t}\gamma_{Y,t}dN_t\end{aligned}$$

Algebra Rules (“Complete”)

Differential Multiplication	Result
dt	dt
dW_t	dW_t
dN_t	dN_t
$(dW_t)^2$	dt
$(dW_t)^m, m \geq 3$	0
$dt dW_t$	0
$(dt)^\alpha (dW_t)^m, \alpha > 0, m \geq 1$	0
$(dN_t)^m, m \geq 1$	dN_t
$dt dN_t$	0
$(dt)^k (dN_t)^m, k \geq 1, m \geq 1$	0
$dN_t dW_t$	0
$(dt)^k (dN_t)^m (dW_t)^n, k \geq 1, m \geq 1, n \geq 1$	0

Lecture Summary

- Introduced Brownian motion and Poisson processes
- Introduced the stochastic integral
- Learnt Itô's lemma
- Learnt stochastic “differential” algebraic rules

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